

Targeting a challenging position as a Senior Developer specializing in C++ and Qt, aiming to contribute to innovative software solutions.

PROFILE SUMMARY

- Accomplished **software engineer with over 2 years of experience**; showcasing proficiency in **C++ and Qt/QML application development**, specializing in creating high-performance, scalable software solutions that drive innovation and enhance user accessibility.
- Proficient in **real-time 3D graphics design utilizing DirectX12 and OpenGL**, with a strong focus on optimizing performance and enhancing visual fidelity for interactive applications.
- Showcased expertise in **multithreading techniques and performance tuning**, resulting in significant **improvements in application efficiency and resource management** across diverse projects.
- Strong foundation in **database management with MySQL and PostgreSQL**, coupled with hands-on experience in **full-stack development using the MERN stack**, ensuring robust data management and dynamic web applications.
- Proven capability to work in **Agile environments**, collaborating effectively with cross-functional teams to **gather requirements, design architectures, and deliver high-quality software solutions** that meet user needs and business objectives.

CORE COMPETENCIES

C++ Development | Qt/QML Application Development | Front-end & Back-end Development | Graphics Programming | Multithreading | Database Management | Software Development Lifecycle (SDLC) | Embedded GUI Development | Version Control | SCRUM | Real-time 3D Graphics Design | Agile & Waterfall Methodologies | Object-oriented Programming | Object Oriented Analysis Design | Testing & Debugging | DSA (Data Structures & Algorithms) | Problem-solving

TECHNICAL SKILLS

Programming Languages	C++ 11, 14 & 17, C, VC++, MFC, SQL, Java & JavaScript
Frameworks and Libraries	Qt, QML, React.js, STL, ATL, DirectX & OpenGL
Databases	MySQL & PostgreSQL
Web Technologies	HTML, CSS, XML & MERN stack (MongoDB, Express.js, React.js, Node.js)
Tools and IDEs	Visual Studio, VS Code, Qt Creator & Git

WORK EXPERIENCE

Scanning Pens Pvt. Ltd., Mumbai | Senior Developer (C++ & Qt) | Oct’23 – Jun’24

Project Undertaken: Scanning Pen

Project Description: Developed and enhanced a handheld text-to-speech scanning pen that integrates optical scanning and TTS technology to assist individuals with reading difficulties, such as dyslexia.

Technologies Used: C++/C, Qt, QML, STL, DSA, SSML (Speech Synthesis Markup Language), TTS (Text to Speech), Git, Agile Methodology, JIRA, Embedded GUI Development using Qt/QML

Key Result Areas:

- Collaborated with cross-functional teams to **gather requirements & translate them into technical specifications** for software development.
- Engaged in **comprehensive software design and architectural planning**, contributing to the creation of scalable and efficient applications.
- Managed development of **maintainable code utilizing modern C++ standards**, ensuring adherence to best practices and industry guidelines.
- Administered **version control systems**, ensuring seamless collaboration and integration of code changes within the development team.
- Conducted **rigorous testing & debugging processes** to mitigate software issues, enhancing overall application reliability & performance.
- Implemented solutions to **optimize existing codebases**, resulting in **improved application performance & reduced resource consumption**.
- Developed and implemented a **user-friendly interface for the Scanning Pen**, enhancing accessibility for users with reading challenges.

Highlights:

- Contributed to **embedded GUI development using Qt/QML**, optimizing performance and **reducing load times by 25%**.
- Resolved complex issues** that enhanced user satisfaction & **improved efficiency by 20%**, boosting product usability & reliability.

Project Undertaken: Drone Designer

Project Description: A real-time 3D TV graphics software designed for WASP3D workflow, featuring an intuitive interface for seamless 3D graphic tasks.

Technologies Used: C++, STL, ATL, Multithreading, DSA, DirectX12, MS-XML, Git & Windows

Key Result Areas:

- Played a key role in the **development of "Drone Designer," a real-time 3D TV graphics design software.**
- **Managed code workflow** effectively, **addressing software bugs** and implementing new features.
- Showcased expertise in **object-oriented programming, shape manipulation, animation, and DirectX12 resources.**
- Enhanced **data storage** and increased memory efficiency to ensure high-performance applications.
- **Streamlined and optimized code modules** for improved efficiency and cohesion.
- **Resolved crashes** and **improved application stability** to enhance the user experience.
- **Collaborated with cross-functional teams** to ensure seamless feature integration.
- Conducted **comprehensive testing and debugging** to uphold high standards of software quality and performance.

Highlights:

- Implemented features that **reduced memory usage by 15%** and enhanced system stability.
- Worked extensively with **DirectX12 in multithreaded environments** to **improve the rendering pipeline's efficiency**, achieving a smoother user experience and **reducing latency by 30% under high workloads.**
- Contributed to a **10% reduction in crashes and overall system performance improvements.**

INTERNSHIP

Avllo Private Limited, Varanasi | Web Developer Intern | Jan'22 – Jun'22

Project Undertaken: Havenwork Dashboard System

Technologies Used: MERN stack (MongoDB, Express.js, React.js, Node.js), Git & Windows

Description: Developed user-friendly interfaces for the Havenwork dashboard and other feature-rich web applications, enhancing usability and overall user experience.

Key Result Areas:

- **Directed multiple projects**, such as the "Havenwork" dashboard, a Slack clone, and a To-Do App, showcasing expertise in web application development.
- Employed innovative **front-end techniques to develop visually engaging and responsive designs.**

EDUCATION

- **2022:** Master of Computer Applications (MCA) from School of Management Sciences, Varanasi, Uttar Pradesh.
- **2020:** Additional Subject: Math from Uttar Pradesh Board.
- **2019:** Bachelor of Computer Applications (BCA) from Indira Gandhi National Open University (IGNOU), New Delhi, Delhi.

CERTIFICATIONS

- Git and Github Essentials from Amigoscode ([Certificate](#))
- Qt Fundamentals from Qt Company
- Agile Foundation from LinkedIn Learning ([Certificate](#))

PERSONAL DETAILS

Address: Noida – 201301, Uttar Pradesh
Date of Birth: 20th September 1996
Languages Known: English & Hindi